OfferPilot

Agentic offer-decision system for e-commerce | Project status report

As of 22 Oct 2026 | Solo hackathon build | github.com/SamiyaS5904/Hackathon-PS_2

OVERALL PROGRESS
2 of 9 build sessions done

22%

The idea. An AI agent (Claude Opus) reads each customer's history and decides **which offer to send, when, or nothing at all**. Plain rule code checks every decision so the company **never loses profit**. Offers go out through one Telegram bot; a company dashboard shows the marketing manager **why** each offer was sent. Business goal: keep customers' ordering habit alive, win back lapsing customers with offers on their favourite items.

DONE SO FAR

- **Planning & docs:** CLAUDE.md (project constitution, approved), PLAN.md, reasoning.md (why/how), TESTING.md (tester guide), documentation.md (build log).
- **Git:** repo on GitHub, 21 small, clear commits so far.
- **Config:** offer building blocks (6 offer types) and default guardrail rules in YAML.
- **Database:** 17-table SQLite schema: company data, hidden truth, pipeline tables.
- **Synthetic data generator:** 8 hidden personas, festivals, paydays, habits, carts, wallet, points, past campaigns with random holdout groups. Full run verified: 0 broken links.

KEY DECISIONS

- **Scope:** only the decide-and-send layer. No customer UI, shop, bargaining, phone numbers.
- **Claude API used only by the agent**; everything else is plain code and works without a key.
- **Model:** Claude Opus. **Plan your day:** manager sets AI decisions/day (default 50), overflow goes to free fallback rules. Est. cost Rs 6-9 per decision; dataset size doesn't change cost.
- **Profit first:** expected profit = P(buy|offer) x margin after offer - P(buy|no offer) x normal margin, computed by code. Min Rs 20 per offer, margin floor 10%.
- **Editable rules:** marketing team edits caps/budgets on the dashboard; every save is a version; unsafe values refused; pending offers re-checked.
- **Testing:** 2-3 real Telegram testers, re-linkable to different customer profiles.

FEATURE STATUS

Planning docs & CLAUDE.md	100%
Offer + rule configs (YAML)	100%
Database schema	100%
Synthetic data (3,000 customers)	100%
Analyst: customer profiles & habits	0%
Profit math + rules checker	10%
Fallback rules + message templates	10%
Gatekeeper + Plan your day	5%
AI agent (Claude Opus, tools)	5%
Telegram bot + tester linking	0%
Event simulator	0%
Company dashboard + API	0%
Test harness (7 test groups)	0%

Green = built and verified. **Amber** = fully designed in CLAUDE.md, code next. Grey = not started.

IDEATION HIGHLIGHTS

- **Uplift thinking:** skip 'buys anyway' and 'won't buy anyway' customers; target persuadables.
- **Building-block offers:** agent assembles type, amount, product, expiry, send time and message.
- **Approve / repair / reject** loop: rejection reasons go back to the agent (max 2 retries), then fallback.
- **Manager-readable reasoning:** summary, signals used, alternatives rejected, expected impact, why now.
- **Hidden ground truth** lets the harness score the agent against what really works.

DATASET GENERATED (REAL NUMBERS FROM THE FULL RUN)

3,000	280	277,049	1,194,656	325,371	35,785	19,683	49,739
Customers	Products	Sessions	Browse events	Cart events	Orders	Past offers	Points entries

Proof the patterns are real: past offers lifted buying by **+12.2 pts for deal hunters** but only **+0.5 pts for loyal regulars** (they buy anyway, so offers there are wasted); **86% of about-to-leave VIPs** are overdue on their usual order gap; festival days have **58% more orders** (139 vs 88 a day). 12 months of data, 26 past campaigns with random holdout groups, 0 broken links.

ROADMAP (DAY 1 = WORKING MVP, DAY 2 = IMPROVE + TEST)

Session 0	Session 1	Session 2	Session 3	Session 4	Session 5	Session 6	Session 7	Session 8
Plan & approve	Configs + data	Analyst + rules	AI agent	Bot + simulator	MVP dashboard	Test harness	Uplift + polish	Final demo

COMING NEXT: SESSION 2

- **Analyst:** one profile per customer: usual order gap, overdue ratio, favourite items, churn risk, discount dependence, cart, wallet, expiring points, offer fatigue.
- **Profit math + rules checker:** approve / repair / reject against the active rules version.
- **Fallback rules** with the 5 fixed message templates; **gatekeeper** + Plan your day.
- **First tests:** every guardrail gets pass / fail / edge-case tests.

WHAT THE FINAL DEMO WILL SHOW

- Manager plans the day, runs the agent, and reads **clear reasoning** for every offer.
- Offer arrives on a real tester's **Telegram**; re-linking a friend to another profile gives a **different** offer.
- Editing a rule (e.g. electronics 8% to 12%) changes the next offers immediately.
- Harness proves: rules never break, tricks are blocked, agent + rules beats send-to-all, never loses money.